

# Hydrogenation of carbon oxides over promoted Fe-Mn catalysts prepared by the microemulsion methodology

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## Abstract

The microemulsion technology was used in order to improve the preparation of iron-based catalysts for the CO and CO<sub>2</sub> hydrogenation reactions. This technique led to a sample with a higher surface area when compared with a similar solid prepared by conventional precipitation. As a consequence, a higher catalytic activity was obtained.

The hydrogenation of carbon oxides was also performed over promoted iron-manganese catalysts. The preparation of these samples by the microemulsion methodology yielded to homogeneous mixed oxides. Manganese-containing catalysts presented a higher activity towards the formation of hydrocarbons, at the same conversion levels, than the iron counterparts.

These catalysts were promoted with copper, sodium and potassium. Carbon dioxide conversion was favoured by alkaline addition, especially by potassium, due to the promotion of the water-gas shift reaction. Furthermore, alkaline promotion enhanced selectivity towards long-chain products either in CO<sub>2</sub> and CO hydrogenation processes. Addition of copper caused a more facile reducibility of the solids, resulting in a significant increment in both carbon monoxide and carbon dioxide conversion without modifying products selectivity.

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**Keywords:** Iron catalysts; Fischer–Tropsch; Carbon dioxide hydrogenation; K-promoted catalysts; Cu-promoted catalysts; Microemulsion

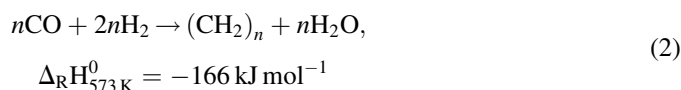
## 1. Introduction

Fischer–Tropsch Synthesis (FTS), consisting of the CO hydrogenation to hydrocarbons, has been an important topic in the last years and it is currently attracting a renewed interest as a key technology for the production of clean transportation fuels and chemicals [1,2]. The products derived from Fischer–Tropsch Synthesis are endowed with a tremendous environmental value due to the lack of harmful substances such as sulphur, nitrogen and aromatic compounds. The starting feedstock of this reaction, the so-called synthesis gas or syngas, is a mixture of H<sub>2</sub> and CO with different H<sub>2</sub>/CO ratios. This mixture can be produced from coal, natural gas or biomass [3]. When coal or biomass are used as source, the resulting syngas contains large amounts of carbon dioxide, thus demanding expensive purification steps (about 5\$/MT CO<sub>2</sub>) [4] which increase significantly the process expenses.

Unlike CO hydrogenation, the understanding of carbon dioxide hydrogenation process is still a major challenge. There have been various attempts to transform carbon dioxide into hydrocarbons, mainly using those catalysts that have been proved active in the Fischer–Tropsch Synthesis, such as Ni, Ru and Co. Most attempts yielded methane as the only product, even if water-gas shift active promoters were added [5]. It seems likely that carbon dioxide hydrogenation proceeds *via* a two-step reaction mechanism [6]. In the first step, carbon dioxide is converted into carbon monoxide through the reverse water-gas shift reaction:



The carbon monoxide produced in this way reacts subsequently according to the Fischer–Tropsch Synthesis producing mainly hydrocarbons:



Some authors have also proposed a mechanism which implies direct carbon dioxide hydrogenation [7,8]. Such reaction

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pattern would only apply for processes carried out under relatively severe conditions (temperature over 673 K). Under reaction conditions similar to those employed in the present work, CO<sub>2</sub> conversion into hydrocarbons occurred mainly *via* CO intermediates as proved by using isotopic-marked carbon dioxide in the feed [9].

Iron oxides have been used as Fischer–Tropsch (FT) catalysts for many years [10,11]. Besides, they are active in both water-gas shift and reverse water-gas shift reactions [12]. In principle, iron-based catalysts would be ideal candidates to be used in the Fischer–Tropsch Synthesis of CO<sub>2</sub>-containing syngas feeds [1,13]. Moreover, iron catalysts are attractive because of the highly olefinic nature of the obtained products which allows their use as a feedstock for chemical industry [14]. However, pure iron catalysts have some drawbacks like a high selectivity to non-desired products such as methane [15] or rapid deactivation. Therefore, promoters must be added in order to improve their stability and to tune the selectivity.

It has been reported that manganese, as a promoter for iron catalysts, enhances the activity of iron-based FT catalysts [16,17], particularly in producing short-chain olefins [18–21]. The catalytic behaviour of mixed iron-manganese oxides was found to be influenced by the preparation technique and the structural properties of the catalytic precursors [22,23].

Alkaline elements are used as promoters because they can modify the adsorption pattern of the reactants (H<sub>2</sub> and CO) on the active sites. The overall effect of potassium on the behaviour of iron-based FTS catalysts, i.e., CO chemisorption enhancement, has been justified as a consequence of the iron tendency to withdraw electronic density from potassium, resulting in a strengthening of the Fe–CO bond [24]. The enhancement in carbon monoxide chemisorption leads to the increment of the selectivity towards olefins and high molecular products along with the depleting of methane formation [25]. Nearly all commercial iron-based catalysts contain potassium amongst several other promoters. When potassium loading is high, CO dissociation proceeds faster than carbon hydrogenation, leading to an excessive carbon deposition that eventually deactivates the catalyst surface. Jiang et al. [26] investigated the effect of potassium on Fe–Mn catalysts by *in situ* Diffuse Reflectance Infrared Fourier Transformed Spectroscopy (DRIFTS). Their results indicated a moderate interaction between potassium and surface iron species that promotes iron oxide reduction to form fine metallic iron clusters. Although potassium is the most common alkaline promoter used, there have been some attempts to study the effect of other alkaline and alkaline-earth elements [5,27,28]. However, the difference amongst alkaline promoters has not been satisfactorily explained yet. Although the effect on the FT performance by potassium promotion of iron-based catalysts has been extensively investigated, not many papers have been published dealing with the effect of potassium on the manganese-promoted precipitated iron catalysts [24]. It has been reported that the addition of alkali metals to Fe–Mn catalysts improves the selectivity to alkenes. Therefore, it is assumed that the interaction between surface iron and the surface manganese species is somehow altered by the presence of potassium [28].

Copper is usually added to iron-based Fischer–Tropsch catalysts because of its ability to enhance hematite reducibility [29]. When copper oxide is reduced to metallic Cu, the crystallites formed provide H<sub>2</sub> dissociation sites, which in turn lead to reactive hydrogen species that are capable to reduce Fe oxides at lower temperatures.

The method of catalyst preparation plays an important role in the physical properties and catalytic performance of the catalysts. In this work, the microemulsion technology has been chosen as preparation method in order to obtain a series of Fe–Mn oxides displaying a homogeneous distribution of both elements throughout the solid. A microemulsion is an optically transparent and thermodynamically stable dispersion of a water phase into an organic phase stabilized by a surfactant [30]. If the minority phase is the aqueous one, then reversed micelles are obtained. In the last years, it has been generalized the use of microemulsions as nanoreactors in order to synthesize nanoparticles of controlled size [31,32].

Amongst advantages such as rendering nanosized particles, usually displaying high surface area and low microporosity, microemulsion is an ideal technique for the preparation of materials containing two (or more) metallic or oxide phases. The different species (oxide precursors) are homogeneously mixed within the micelles, therefore rendering solids displaying high internal homogeneity and an optimal interaction between its constituents [32]. Some authors have prepared supported iron-based Fischer–Tropsch catalysts by microemulsion, reporting high activity and marked selectivity to oxygenates [33]. However, there are still few works dealing with the preparation of non-supported catalysts even though the synthesis of mixed ferrites by microemulsion has been reported [34,35].

In the present work, the catalytic performance of Fe–Mn samples in the CO<sub>2</sub> hydrogenation reaction was studied. The expected path for this reaction is the following: (i) water-gas shift reaction to produce CO and (ii) CO hydrogenation. Therefore, the performance of the samples in the CO hydrogenation reaction was previously evaluated. We have found that the extension of the CO<sub>2</sub> hydrogenation accounts for both water-gas shift and CO hydrogenation processes.

The influence of some promoters, such as Na, K and Cu, on the catalytic performance of Fe–Mn mixed oxides in the hydrogenation of carbon monoxide and carbon dioxide has been also studied.

## 2. Experimental

### 2.1. Catalysts preparation

The Fe–Mn catalyst precursors were prepared by coprecipitation in a water-in-oil microemulsion. A water solution of metal precursors, Fe(NO<sub>3</sub>)<sub>3</sub>·9H<sub>2</sub>O (Fluka puriss, p.a. ACS: 98–100%) and/or Mn(NO<sub>3</sub>)<sub>2</sub>·4H<sub>2</sub>O (Fluka, purum, p.a. >97%) was added to a mixture of an oil phase (isooctane, Aldrich, >99%) and the commercial available non-ionic surfactant Tergitol 15-S-5, which consists in a secondary alcohol ethoxylate {C<sub>12–14</sub>H<sub>25–29</sub>O(CH<sub>2</sub>–CH<sub>2</sub>O)<sub>x</sub>} with hydrophilic-to-lipophilic

balance (HLB) 10.5 and molecular weight of *ca.* 413 g mol<sup>-1</sup>. The resulting mixture had the following composition: 10 wt.% of aqueous phase, 70 wt.% of oil phase and 20 wt.% of surfactant. Besides, a small amount of ethanol (maximum 3 wt.%) was added in order to obtain a proper microemulsion when needed. The role of the ethanol addition is to increase the microemulsion stability [32]. The presence of a cosurfactant is required when the ionic concentration inside the micelles is high, because in this case, the electrostatic interactions between the water molecules and the polar part of the surfactants are debilitated and the interface become more rigid [37,38].

After stirring, a transparent mixture, stable for at least 24 h, was obtained. A similar microemulsion containing NH<sub>4</sub>OH (Panreac, Assay 28.0–30.0%) in the aqueous phase was used as the precipitating agent. The resulting mixture was let to decant overnight. The solid was recovered by centrifugation and washed thoroughly with distilled water and ethanol. Finally, the samples were dried overnight at 383 K and subsequently calcined in air at 773 K for 6 h (4 h in the case of samples subjected to further metal promotion). Potassium, sodium and/or copper promoters were added by wet impregnation with the carbonate (Na and K) or nitrate (Cu) precursors. K, Na and Cu loadings were set to the optimal value according to Li et al. [39]. The promoted catalysts were dried at 383 K for 16 h and calcined at 773 K for 2 h in air. Besides, a pure iron oxide (*p*-100Fe) catalyst prepared by precipitation of iron nitrate with NH<sub>4</sub>OH was used as a reference material [36]. The labelling, composition and specific surface area values of the calcined samples are summarized in Table 1.

## 2.2. Characterization

Powder X-ray diffraction (XRD) patterns were recorded in the 5–95° 2θ range in the scan mode (0.05°, 2 s) using a Seifert 3000 XRD diffractometer equipped with a PW goniometer with Bragg–Brentano θ/2θ geometry, an automatic slit, and a bent graphite monochromator.

Temperature programmed reduction (TPR) experiments were carried out in a Micrometrics 3000 apparatus. The sample (20–35 mg) was pretreated under dry air at 383 K for 1 h. The TPR profiles was recorded by heating the sample from room temperature to 1173 K at a rate of 10 K min<sup>-1</sup>, under a 10 vol.% H<sub>2</sub>/Ar flow. The H<sub>2</sub> consumption was monitored with a thermal

conductivity detector (TCD). In order to avoid water condensation from the exit stream, the effluent gas was passed through a cold trap placed before the TCD.

SEM-EDS analyses were performed on a Philips XL30 scanning electron microscope coupled to an EDAX DX4i analyser for energy-dispersive X-ray analysis at 20 keV to obtain quantitative information on the distribution of Fe and Mn atoms. The powder samples were sputter coated with gold with a Sputter Coater SC502 previously to the analysis.

Nitrogen adsorption isotherms were recorded at the temperature of liquid nitrogen (77 K), using a Micromeritics ASAP 2000 apparatus. The samples were degassed at 413 K for 12 h prior to the determination of the adsorption isotherm.

## 2.3. Catalytic performance

Catalysts were tested in CO<sub>2</sub> and CO hydrogenation reaction using a fixed-bed microreactor. Flow rates were controlled using Bronkhorst High-Tech Series mass flow controllers. The calcined samples (110 mg, 0.25–0.30 mm particle size) were diluted with SiC (*ca.* 2.5 g, 0.25–0.30 mm particle size) to avoid hot spots. First, the catalysts were activated *in situ* at 553 K for 1 h in syngas (31CO/62H<sub>2</sub>/7N<sub>2</sub>, GHSV = 0.0023 L g<sup>-1</sup> s<sup>-1</sup>) at atmospheric pressure. The reactor was then cooled down in synthesis gas to the starting reaction temperature (518 K) and pressurized at 1.01 MPa. CO hydrogenation was measured at 518, 548 and 573 K. Then, the feed was switched to a mixture 23CO<sub>2</sub>/69H<sub>2</sub>/8N<sub>2</sub> (GHSV = 0.0023 L g<sup>-1</sup> s<sup>-1</sup>). CO<sub>2</sub> hydrogenation activity was measured at 573 K for 48 h.

Analysis of reactant gases and products was performed *on-line* with a gas chromatograph (Varian CP-3800). A polymeric column connected to a thermal conductivity detector was used to analyse inorganic gases (H<sub>2</sub>, N<sub>2</sub>, CO and CO<sub>2</sub>) and water. Hydrocarbons and oxygenated compounds were analysed with a Restek Rtx-1 capillary column (105 m × 0.25 mm × 0.25 μm) connected to a flame ionization detector (FID). N<sub>2</sub> was used as internal standard for chromatographic analyses.

A more detailed description of the reaction unit has been published elsewhere [36].

## 3. Results and discussion

### 3.1. Catalyst characterization

The XRD analyses of the non-calcined samples (data not shown) displayed broad peaks indicative of amorphous solids, resembling that of the 2-line ferrihydrite [40]. Fig. 1 shows the XRD patterns of the calcined samples. The mean particle size of each sample, calculated by applying the Debye–Scherrer equation, is also depicted. In all cases, reflections ascribed to the hematite iron oxide structure (JCPDS 84–307) were the only ones observed. Reflections from isolated manganese oxide phases were not detected in the bimetallic Fe–Mn samples, probably as a result of the incorporation of manganese into the hematite lattice, in good agreement with previously reported results [41–43]. On the other hand, we have observed that the lattice parameters of hematite (data not shown) for the

Table 1  
Nomenclature, surface area, theoretical (atomic percentage) and experimental (in brackets) catalysts composition obtained from EDX analyses

| Nomenclature                 | Surface area<br>(m <sup>2</sup> g <sup>-1</sup> ) | Fe:Mn<br>atomic ratio | K<br>(at.%) | Na<br>(at.%) | Cu<br>(at.%) |
|------------------------------|---------------------------------------------------|-----------------------|-------------|--------------|--------------|
| <i>p</i> -100Fe <sup>a</sup> | 14                                                | 100:0                 | 0           | 0            | 0            |
| 100Fe                        | 18                                                | 100:0                 | 0           | 0            | 0            |
| 95Fe                         | 25                                                | 95:5                  | 0           | 0            | 0            |
| 95Fe–Cu                      | 25                                                | 95:5                  | 0           | 0            | 1.0 (1.1)    |
| 95Fe–K                       | 18                                                | 95:5                  | 1.3 (0.5)   | 0            | 0            |
| 95Fe–Na                      | 30                                                | 95:5                  | 0           | 1.3          | 0            |
| 95Fe–CuK                     | 23                                                | 95:5                  | 1.3 (0.9)   | 0            | 1.0 (1.1)    |

<sup>a</sup> Catalyst prepared by conventional precipitation.

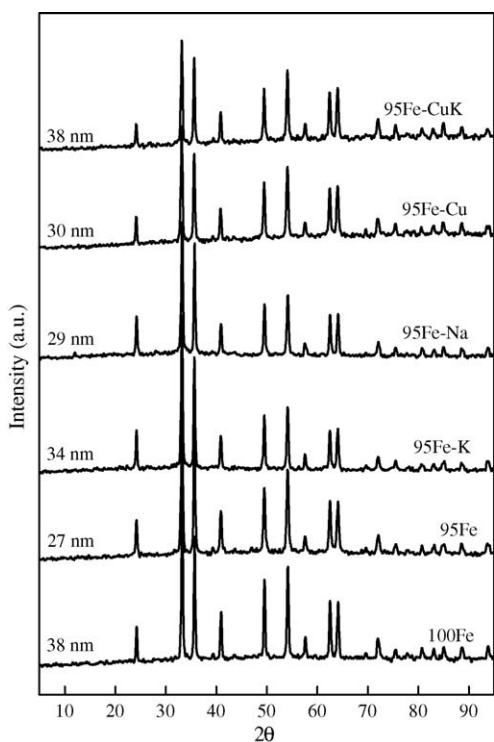


Fig. 1. XRD patterns of calcined samples and mean particle size calculated from the Debye-Scherrer equation.

iron-manganese samples did not change significantly when compared to those of the Fe samples. This observation is not in disagreement with the incorporation of manganese into the hematite structure because hematite lattice can incorporate up to 15 at.% of manganese [44].

In the promoted samples, neither alkaline nor copper species were detected by XRD, probably because of their low concentration. The crystallinity of the samples was not modified by the presence of the promoters. This result is not unexpected since they were impregnated over the Fe-Mn mixed oxides once they were calcined at 773 K for 2 h. After this treatment, the hematite structure is known to remain stable [29].

The particle size of the manganese-free sample was *ca.* 38 nm, decreasing to *ca.* 27 nm when manganese was added to the sample.

The specific surface area values of calcined samples are collected in Table 1. BET surface area values ranged from 14 to 30 m<sup>2</sup> g<sup>-1</sup>. The iron-manganese oxides displayed a higher specific area than the pure iron sample. This feature is the result of the smaller crystal size of the iron-manganese samples compared to the size of the pure Fe sample. Also, it has been reported that doping agents can inhibit iron sintering during the calcination process [29]. The addition of copper after the calcinations process did not modify the surface area of the solids. The effect of the addition of alkaline promoters on the surface area is ambiguous. Whereas, as already reported, potassium caused a slight decrease in the specific area values of the samples [24] sodium addition increased the surface area of the samples, probably due to the small size of sodium oxide entity [45].

Fig. 2 shows the hydrogen consumption profiles of the calcined samples. In all cases, two reduction peaks corresponding to the sequential transformation of Fe<sub>2</sub>O<sub>3</sub> (hematite) to Fe<sub>3</sub>O<sub>4</sub> (magnetite) and subsequent formation of metallic Fe were observed. The ratio of the areas under the two peaks was 8.4 in the case of pure iron oxide (100Fe), in fair agreement with the theoretical value of 8.0 if a two-stage reduction mechanism is assumed [6]. For the 95Fe samples, the ratio decreased to *ca.* 7.4. This feature accounts for the simultaneous reduction of Fe<sup>3+</sup> and Mn<sup>3+</sup> to Fe<sup>0</sup> and Mn<sup>2+</sup>, respectively, within the hematite lattice at this step [22]. Thus, the area of the first peak is higher when compared to that of the pure Fe sample. By comparing the TPR profiles of samples 100Fe and 95Fe, it can be observed that manganese did not affect the reduction pattern of hematite. However, a shoulder at *ca.* 800 K can be observed for the sample 95Fe. A peak at this temperature has been already observed and ascribed to the reduction of the manganowustite mixed oxide, Fe<sub>x</sub>Mn<sub>1-x</sub>O to Fe<sup>0</sup> and MnO [6,42]. The reduction of the sample 95Fe ends at a higher temperature because, as reported by other authors, manganese retards the reduction of iron oxides [16]. From these results, a further evidence of the formation of mixed iron-manganese phases can be inferred.

Both potassium and sodium promoters show no effect in the reduction behaviour of the hematite (see 95Fe, 95Fe-K and 95Fe-Na in Fig. 2). This is in contrast with the results published by Li et al. [29], who reported that a higher temperature is required for metallic iron formation in a K-promoted Fe<sub>2</sub>O<sub>3</sub> sample.

On the other hand, the addition of Cu to 95Fe led to a 150 K shift of the reduction process temperature towards lower values,

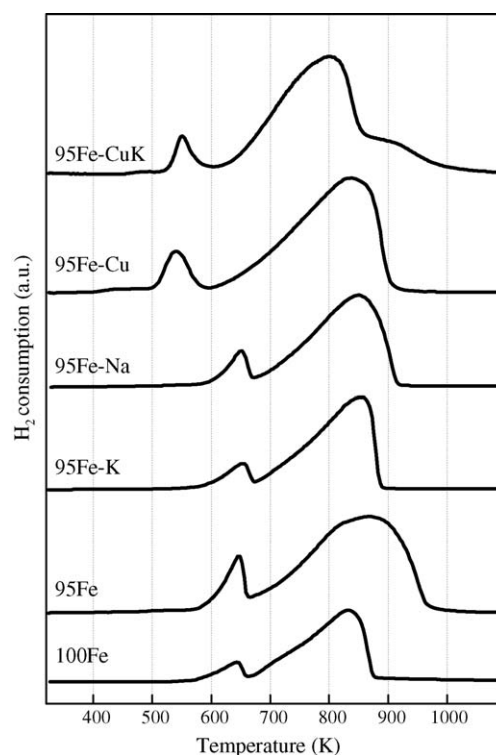


Fig. 2. TPR profiles of calcined samples.

probably as a result of the presence of sites for  $H_2$  dissociation formed by reduction of  $CuO$  to  $Cu$  metal at low temperatures. In all cases, the theoretical  $H_2$  consumption (calculated taking into account the weight of sample used) agrees relatively well with the experimental values. The addition of promoters does not change the amount of  $H_2$  required for the sample reduction. We have just observed that  $Cu$  addition shifts the reduction temperature to lower values, remaining unaltered the amount of  $H_2$  needed for the reduction process.

SEM-EDX analyses were carried out in an attempt to study the morphology of the catalyst precursors. Backscatter ad-images obtained from scanning electron microscopy analyses are depicted in Fig. 3. The images of the non-promoted samples revealed similar morphologies. The EDX-analyses were obtained in the same solids as those shown in the pictures

and the results are collected in Table 1. It can be seen that preparation of oxides by microemulsion resulted in a homogeneous distribution of the components, either at low or long range magnifications (5000 $\times$ ). The surface of the samples remained unaltered after copper addition. However, potassium-promoted samples showed surface agglomerates of non-defined shape. EDX analysis of these surface agglomerates revealed that their composition was poor in manganese. When copper was added, the surface composition was a combination of that of the precursors plus that of copper. Micrographs of sodium-promoted samples were not recorded because this element could not be detected due to the limitations of EDX technique [46]. The difference in the potassium content between the theoretical and the experimental values is also related to some limitations of the EDX technique.

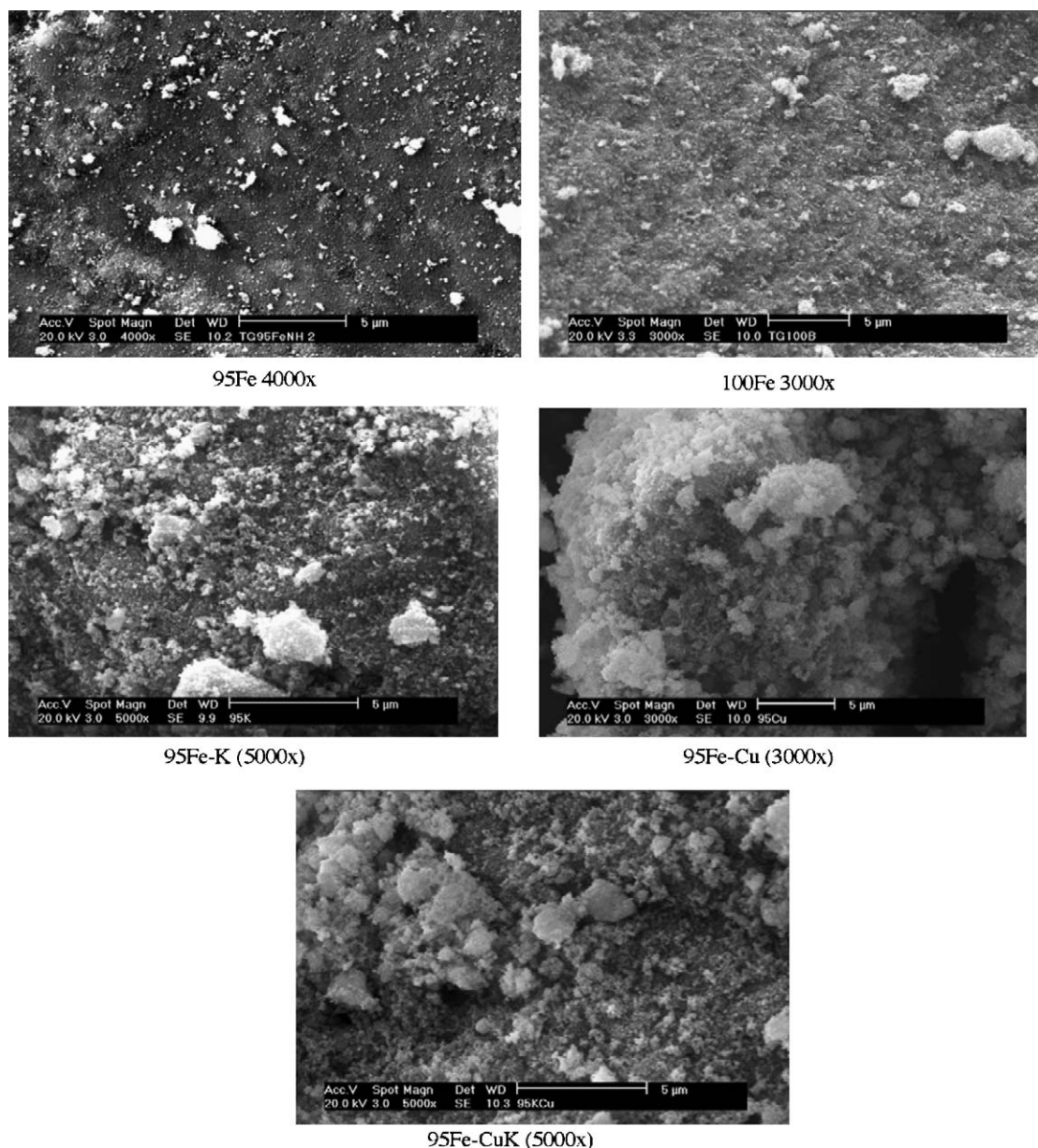


Fig. 3. SEM micrographs of calcined samples.

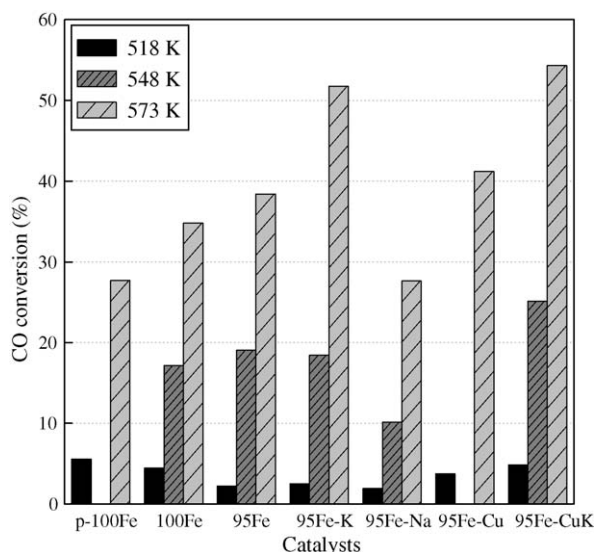


Fig. 4. CO conversion with the Fe-Mn catalysts at different reaction temperatures.

### 3.2. Catalytic performance in CO hydrogenation

Before measuring the catalytic activity, preliminary tests were conducted to ensure the absence of mass transfer limitation phenomena. Under the studied conditions, carbon monoxide hydrogenation rate increased exponentially with temperature for all samples as shown in Fig. 4. However, the magnitude of this increment depended on the catalysts composition. The pure iron catalyst (100Fe) showed the lowest activation energy value amongst all the samples ( $E_a = 96 \text{ kJ mol}^{-1}$ ), which is in good agreement with the literature [47]. The mixed iron-manganese sample 95Fe displayed the highest activation energy value ( $163 \text{ kJ mol}^{-1}$ ) of the series. However, promotion with copper reduced it to values close to that of 100Fe.

Fig. 5 depicts the hydrocarbon production rate of the different catalysts, expressed as grams of hydrocarbon monomer unit ( $-\text{CH}_2-$ ) per gram of catalyst per second at 518, 548 and 573 K.

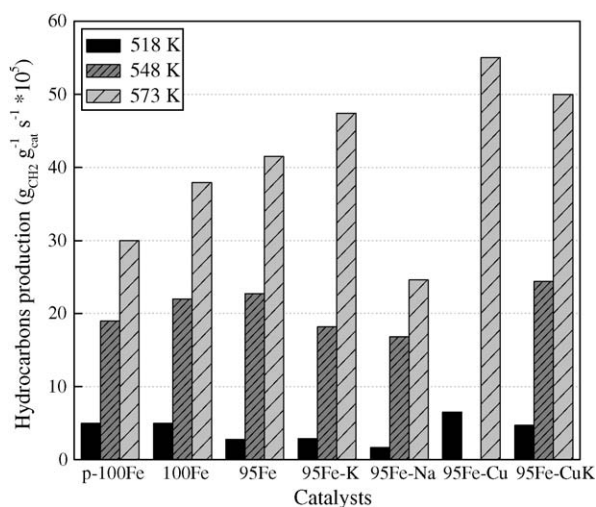


Fig. 5. CO hydrogenation rate to hydrocarbons with the Fe-Mn catalysts at different reaction temperatures.

By comparison of 100Fe versus *p*-100Fe, it can be seen that the microemulsion technology led to a sample displaying a higher catalytic activity. This increase is probably related to the higher surface area of the 100Fe catalyst (Table 1).

The role of Mn depended upon the reaction temperature. While at lower reaction temperature it inhibits the production of hydrocarbons, at 573 K the opposite trend was found (see Table 2). This feature accounts for the fact that Mn retards the overall reduction process of the solids as discussed before. Therefore, higher reaction temperatures can overcome it. Carbon dioxide production was found to follow a similar trend. It was inhibited by the presence of Mn at low reaction temperatures (518 K), but it was slightly promoted at higher reaction temperatures. Nonetheless, the effect of Mn was also found to be dependent on the reaction temperature. Similar olefin-to-paraffin (o/p) values were found for samples either containing Mn or not (95Fe and 100Fe); the largest differences being observed when the reaction was conducted at 573 K. Interestingly, the selectivity towards the  $\text{C}_5\text{--C}_{12}$  hydrocarbons increased with the temperature for sample 95Fe while the opposite trend was found for sample 100Fe.

The effect of Cu promotion is less clear. On the one hand, copper-enhanced hydrocarbon production in the whole range of studied temperatures, probably due to the more facile reduction of the Cu-promoted samples. On the other hand, both chain growth probability and o/p ratio were found unaltered by the presence of Cu, especially at 573 K.

Alkaline promotion, either by Na or K, played a marked role in the performance of the 95Fe samples. The presence of Na rendered less active samples, i.e., sodium had a negative effect on the CO conversion level at all temperatures. Furthermore, hydrocarbons ( $-\text{CH}_2-$ ) production rate was also reduced, even to a larger extent. This can be explained by the enhancement of the  $\text{CO}_2$  production, especially at low reaction temperatures. The effect of K was somehow different. While at low temperatures similar CO conversions were found (samples 95Fe versus 95Fe-K), the CO conversion rate increased over the K-containing samples with increasing the reaction temperature. The  $\text{CO}_2$  production was larger over the K-containing samples (95Fe-K), reaching similar values to those found over the sample 95Fe-Na. Besides, alkaline promoted samples yielded the higher o/p ratios of the series. Whereas the role of Na was more evident with the increasing reaction temperature, the trend was the opposite for the potassium promoted samples, being its effect more evident at low reaction temperatures. The nature of the olefin promotion (increasing value of the o/p ratio) of alkaline agents is ascribed to their capacity to titrate acid sites on the catalysts surface which are responsible for secondary reactions [29].

Under the conditions reported in this paper, copper-promoted samples displayed the best results of the series in terms of hydrocarbon production. This would be due to the fact that iron catalysts promoted with Cu reached a steady state with more carbide species than the Cu-free samples [48]. When copper is reduced to its metallic state at the pretreatment temperature (553 K), it could provide active sites for hydrogen dissociation, promoting hematite reduction to magnetite.

Table 2  
Catalytic performance of Fe-Mn catalysts in the CO hydrogenation

| Catalyst        | $X_{\text{CO}}^{\text{a}}$<br>(%) | $\text{HC}^{\text{b}}$<br>( $\text{g}_{\text{CH}_2} \text{ g}_{\text{cat}}^{-1} \text{ s}^{-1}$ ) | Products selectivity (%) |                                      |                 |                                |                                 |                              | o/p <sup>c</sup> | $\alpha^{\text{d}}$ |
|-----------------|-----------------------------------|---------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------|-----------------|--------------------------------|---------------------------------|------------------------------|------------------|---------------------|
|                 |                                   |                                                                                                   | CO <sub>2</sub>          | Hydrocarbons (CO <sub>2</sub> -free) |                 |                                |                                 |                              |                  |                     |
|                 |                                   |                                                                                                   |                          | Oxygenates                           | CH <sub>4</sub> | C <sub>2</sub> –C <sub>4</sub> | C <sub>5</sub> –C <sub>12</sub> | C <sub>13</sub> <sup>+</sup> |                  |                     |
| <i>p</i> -100Fe | 27.7                              | 27.4                                                                                              | 20.9                     | 14.6                                 | 19.4            | 31.7                           | 30.3                            | 4.0                          | 0.9              | 0.59                |
| 100Fe           | 34.8                              | 37.9                                                                                              | 20.2                     | 0.6                                  | 17.5            | 47.5                           | 32.7                            | 1.8                          | 0.8              | 0.66                |
| 95Fe            | 38.4                              | 41.5                                                                                              | 21.3                     | 1.9                                  | 18.1            | 48.0                           | 30.8                            | 1.3                          | 1.2              | 0.62                |
| 95Fe-K          | 51.8                              | 47.4                                                                                              | 31.1                     | 1.0                                  | 14.2            | 48.4                           | 34.9                            | 1.5                          | 1.2              | 0.62                |
| 95Fe-Na         | 27.6                              | 24.6                                                                                              | 33.6                     | 1.5                                  | 10.1            | 40.8                           | 40.6                            | 7.1                          | 2.0              | 0.64                |
| 95Fe-Cu         | 41.2                              | 55.1                                                                                              | 21.8                     | 1.3                                  | 17.7            | 47.6                           | 31.1                            | 2.3                          | 1.1              | 0.62                |
| 95Fe-CuK        | 54.3                              | 50.1                                                                                              | 34.5                     | 1.7                                  | 14.5            | 46.9                           | 31.8                            | 5.1                          | 1.3              | 0.63                |

Reaction conditions: 573 K, 1.01 MPa,  $\text{H}_2/\text{CO} = 2$  (GHSV =  $0.0023 \text{ L g}^{-1} \text{ s}^{-1}$ ).

<sup>a</sup> CO conversion.

<sup>b</sup> Hydrocarbon production ( $\times 10^5$ ).

<sup>c</sup> Olefin-to-paraffin ratio.

<sup>d</sup> Chain growth probability.

Besides, a larger number of magnetite nuclei are formed in the copper-promoted samples. Therefore, the activation of Cu-promoted samples under syngas yielded more carbide sites [39]. It must be recalled that both  $\text{Fe}_3\text{O}_4$  and  $\text{FeC}_x$  species are formed simultaneously from  $\text{Fe}_2\text{O}_3$  when exposed to syngas.

The transformation of the oxide phase into iron carbide is a process controlled by oxygen diffusion from the bulk to the surface [49]. This process leads to a significant loss in crystalline volume due to the higher density of the carbide crystallites as reported by Li et al. [39,48]. The formation of this external carbide shell inhibits further oxygen diffusion from the bulk, thus stabilizing the bulk magnetite phase.

As discussed above, a large number of small carbide crystallites were formed on the surface the Cu-promoted samples. Due to the large number of small crystallites, the surface of the solid is not a continuous layer, displaying fractures or channels that allow further oxygen diffusion from the bulk. Therefore, the formation of more carbide crystals is not impeded. Thus, at the steady state, copper-promoted catalysts displayed a larger carbide

area, and as a result, more active sites for carbon monoxide adsorption would be available, enhancing the catalytic activity.

The hydrocarbon selectivity distribution (without  $\text{CO}_2$ ) at isoconversion ( $X_{\text{CO}}$ : 2–6%) is depicted in Fig. 6. Manganese promotion enhanced methane production (95Fe versus 100Fe). However, doubly promoted catalysts, displayed the lowest selectivity to methane of the series (see 95Fe, 95Fe-K and 95Fe-Na). Alkaline promoters can reduce the hydrogenation reaction by decreasing the availability of adsorbed hydrogen atoms ( $\text{H}^*$ ) required for the termination of the chain growth *via* hydrogen addition reactions [50,51]. Similarly, the selectivity to heavy fractions was increased over the alkaline promoted catalysts. The selectivity to long-chain products (gasoline fraction  $\text{C}_5\text{--C}_{12}$  and diesel fraction  $\text{C}_{13+}$ ) was enhanced over potassium and sodium promoted samples. Again, the role of sodium as a strong alkaline promoter was evidenced by increasing selectivity towards heavy fractions.

Carbon dioxide production rate in the CO hydrogenation reaction at different temperatures is depicted in Fig. 7. It has

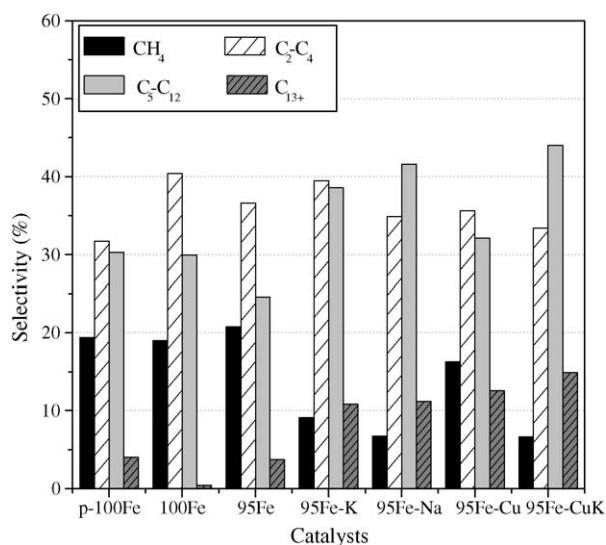


Fig. 6. Hydrocarbon selectivity ( $\text{CO}_2$ -free) at isoconversion ( $X_{\text{CO}}$ : 2–6%).

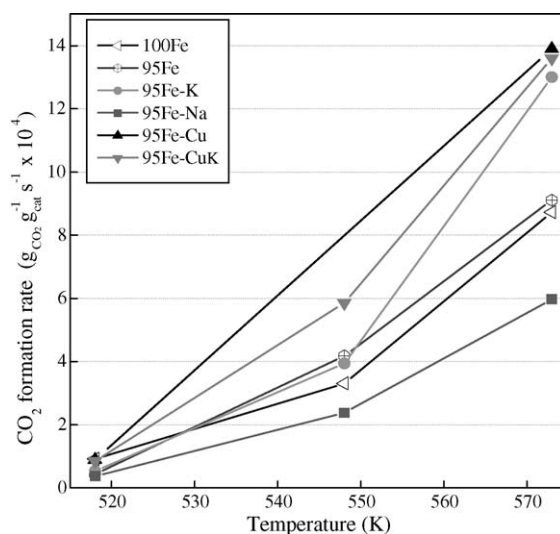


Fig. 7.  $\text{CO}_2$  formation rate vs. the reaction temperature.

been reported that carbon dioxide production rate is proportional to the overall water-gas shift (WGS) reaction rate [52]. The rate of WGS was very low and almost identical for all catalysts at low carbon monoxide conversion levels, probably due to the very low amount of water available to react. The carbon monoxide conversion triggered and the differences in carbon dioxide production rates became evident by increasing the reaction temperature. Except for the Na-containing samples, carbon dioxide production was higher over the promoted catalysts at all CO conversion levels [9]. Potassium-promoted catalysts displayed the highest WGS rate of the series, overcoming that of the non-promoted ones. As suggested by previous studies, manganese addition did not increase the rate of WGS reaction [53]. As discussed above, we have found that the effect of Mn depended upon the reaction temperature, inhibiting CO<sub>2</sub> formation at low reaction temperatures and being negligible at higher reaction temperatures. When it comes to CO<sub>2</sub> production from CO, it seems like that the role of Cu was more important than that of the Mn. For instance, the sample 95Fe-Cu displayed higher carbon dioxide selectivity than 95Fe or 95Fe-K irrespectively of the reaction temperature. This is due to the promoter effect of copper for oxygen removal *via* primary carbon dioxide formation [9] instead of water. The extent of WGS is very low at low temperature, being more important the formation of carbon dioxide *via* the primary mechanism.

An estimate of the extent of the water-gas shift reaction can be obtained by following the WGS reaction quotient (RQ<sub>WGS</sub>) [14]:

$$RQ_{WGS} = \frac{P_{CO_2} P_{H_2}}{P_{CO} P_{H_2O}} \quad (3)$$

where  $P_j$  is the partial pressure of species  $j$  in the effluent stream. The approach to the WGS equilibrium can be described by the parameter  $\eta$ , obtained by dividing the WGS reaction quotient by the equilibrium constant at the reaction temperature:

$$K_{WGS,T} = \left[ \frac{P_{CO_2} P_{H_2}}{P_{CO} P_{H_2O}} \right]_{eq,T} \quad (4)$$

$$\eta = \frac{1}{K_{WGS}} \frac{P_{CO_2} P_{H_2}}{P_{CO} P_{H_2O}} \quad (5)$$

where  $K_{WGS,T}$  is the WGS equilibrium constant at the temperature  $T$ . In this work, the equilibrium constant was calculated from the equation proposed by Huff and Sattelfield [54]. The value of  $\eta$  ranged from 0 to 1, the latter being the equilibrium value [9]. The value of  $\eta$  versus the reaction temperature is represented in Fig. 8. In general, the value of  $\eta$  (and subsequently of the reaction quotient) was small at low carbon monoxide conversions and increased at higher carbon monoxide conversions in good agreement with previous works [52]. This value was higher for the alkaline promoted catalysts, while samples 95Fe and 95Fe-Cu displayed similar values lower than those of the alkaline promoted samples. Nevertheless, the value of  $\eta$  was less than 0.1 in all cases, indicating that the WGS reaction was still far from equilibrium. Therefore, the differences in the carbon dioxide production rate for the different catalysts were due to the

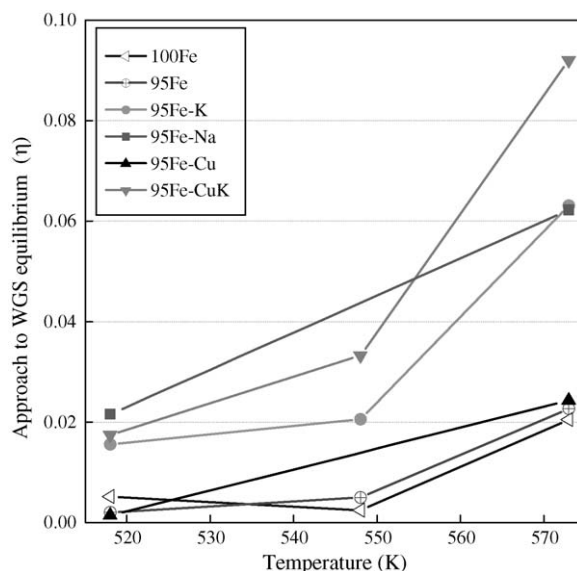


Fig. 8. Approach to the WGS equilibrium vs. the reaction temperature.

promoting effect of basic metals and were not affected by equilibrium limitations of the water-gas shift reaction.

### 3.3. Catalytic performance in CO<sub>2</sub> hydrogenation

Table 3 summarizes the catalytic performance of the promoted and unpromoted iron oxide catalysts in the CO<sub>2</sub> hydrogenation reaction. Carbon dioxide conversion values were lower than the equilibrium ones (75% of CO<sub>2</sub> conversion at 573 K, 1.01 MPa and H<sub>2</sub>/CO<sub>2</sub> = 3). Equilibrium values were calculated by assuming that carbon dioxide hydrogenation to hydrocarbons proceeds in a two-step process [7]. First, CO<sub>2</sub> is converted to CO *via* the reverse water-gas shift reaction (CO<sub>2</sub> + H<sub>2</sub> → CO + H<sub>2</sub>O). Then, CO is transformed into different hydrocarbons according to the classical Fischer–Tropsch mechanism.

As it can be seen in Table 3, the addition of potassium had a positive effect on the carbon dioxide conversion rate (see 95Fe, 95Fe-K and 95Fe-CuK). Also, a certain effect of sodium can be observed. The hydrocarbon production rate was about four times higher with 95Fe-Na than with 95Fe. On the other hand, promotion with copper had no significant effect (see 95Fe-Cu), although the doubly promoted catalyst 95Fe-CuK showed the highest hydrocarbon productivity of the series. As discussed before, Cu promotion showed an important effect on the CO<sub>2</sub> production from CO. However, CO<sub>2</sub> production was attained by primary mechanisms instead of promoting the WGS reaction. Therefore, the reverse reaction, i.e., the CO production from CO<sub>2</sub>, would not be promoted over such systems.

Chain growth probability ( $\alpha$ ) ranged between 0.29 and 0.36 for iron and iron-manganese catalysts, either as prepared or promoted with copper. When potassium or sodium was added, the  $\alpha$  value increased to 0.52 (95Fe-K) and 0.57 (95Fe-Na). The doubly promoted catalyst 95Fe-CuK showed the highest chain growth probability ( $\alpha$  = 0.63). In this case, the synergistic effect of promoters was evidenced [29]. Some authors have reported

Table 3  
Catalytic performance of Fe-Mn catalysts in the CO<sub>2</sub> hydrogenation

| Catalyst | $X_{\text{CO}_2}$<br>(%) <sup>a</sup> | $\text{HC}^{\text{b}}$<br>( $\text{g}_{\text{CH}_2} \text{ g}_{\text{cat}}^{-1} \text{ s}^{-1}$ ) | Products selectivity (%) |                        |                 |                                |                                 |                              | o/p <sup>c</sup> | $\alpha^{\text{d}}$ |
|----------|---------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------|------------------------|-----------------|--------------------------------|---------------------------------|------------------------------|------------------|---------------------|
|          |                                       |                                                                                                   | CO                       | Hydrocarbons (CO-free) |                 |                                |                                 |                              |                  |                     |
|          |                                       |                                                                                                   |                          | Oxygenates             | CH <sub>4</sub> | C <sub>2</sub> –C <sub>4</sub> | C <sub>5</sub> –C <sub>12</sub> | C <sub>13</sub> <sup>+</sup> |                  |                     |
| 100Fe    | 16.6                                  | 9.6                                                                                               | 43.0                     | 0.4                    | 43.2            | 51.7                           | 4.5                             | 0.1                          | 0.7              | 0.36                |
| 95Fe     | 8.1                                   | 2.7                                                                                               | 66.9                     | 0.1                    | 54.9            | 42.7                           | 2.3                             | 0.0                          | 1.0              | 0.29                |
| 95Fe-K   | 18.9                                  | 13.6                                                                                              | 26.9                     | 0.6                    | 28.3            | 54.3                           | 16.0                            | 0.8                          | 0.9              | 0.52                |
| 95Fe-Na  | 9.9                                   | 5.8                                                                                               | 41.2                     | 1.7                    | 27.1            | 53.4                           | 16.9                            | 0.7                          | 1.4              | 0.57                |
| 95Fe-Cu  | 10.9                                  | 6.0                                                                                               | 59.8                     | 0.8                    | 54.9            | 41.8                           | 2.6                             | 0.0                          | 0.8              | 0.30                |
| 95Fe-CuK | 21.8                                  | 18.0                                                                                              | 35.7                     | 2.5                    | 20.2            | 54.1                           | 21.3                            | 2.0                          | 1.4              | 0.63                |

Reaction conditions: 573 K, 1.01 MPa, H<sub>2</sub>/CO<sub>2</sub> = 3 (GHSV = 0.0023 L g<sup>-1</sup> s<sup>-1</sup>).

<sup>a</sup> CO<sub>2</sub> conversion.

<sup>b</sup> Hydrocarbon production ( $\times 10^5$ ).

<sup>c</sup> Olefin-to-paraffin ratio.

<sup>d</sup> Chain growth probability.

that since carbon dioxide hydrogenation follows a reaction mechanism similar to that of carbon monoxide hydrogenation (with a previous CO<sub>2</sub> shift step), the product distribution for both reactions should be very similar when using the same catalyst and reaction conditions [5]. In this work, only the doubly promoted catalyst (95Fe-CuK) displayed comparable values of  $\alpha$  (0.63) for carbon monoxide and carbon dioxide hydrogenation. This fact could be due to the low residence time employed in the reaction tests (*ca.* 0.05 g<sub>cat</sub> s mL<sup>-1</sup>), which did not allow the formation of final reaction products (hydrocarbons) [7].

The selectivity to C<sub>5</sub>–C<sub>12</sub> hydrocarbons (gasoline fraction) increased from 4.5% in the case of pure iron catalyst to 16.9 and 21.3% over 95Fe-Mn and 95Fe-CuK, respectively. This means that the promoters played an important role in the chain growth probability. The selectivity to C<sub>13</sub><sup>+</sup> hydrocarbons (diesel fraction) remained below 2% in all catalysts.

The olefin-to-paraffin ratio increased in the iron-manganese catalysts with respect to the pure iron sample, in parallel with the results obtained from the CO hydrogenation experiments. The potassium-promoted iron-manganese sample (95Fe-K) had the same olefin-to-paraffin ratio value than the non-promoted one. However, the sample 95Fe-Na presented a higher value (1.4 for 95Fe-Na versus 0.9 and 1.0 for 95Fe-K and 95Fe, respectively). This trend was also observed during the CO hydrogenation experiments. The difference between sodium and potassium promotion is mainly reflected in the o/p ratio. Sodium, having more Lewis basic character than potassium, adsorbed carbon monoxide stronger, inhibiting olefin secondary reactions and also, impeding hydrogen chemisorption. This fact also caused the depletion of the catalytic activity [55].

Methane selectivity was higher in the copper-promoted catalysts. Copper favoured hydrogen chemisorption and its subsequent dissociation. Therefore, the surface of copper-promoted catalysts was enriched with hydrogen, favouring methane formation. However, when copper was added together with potassium, the effect of the latter seemed to be predominant and methane selectivity decreased strongly. The same result was obtained when carbon monoxide hydrogenation reaction was carried out (see above).

Our results show that K-promoted catalysts displayed the best catalytic performance in the CO<sub>2</sub> hydrogenation reaction. In this case, the hydrocarbon productivity is at least two times higher than that obtained with the other catalysts here studied. Hydrocarbon formation from carbon dioxide hydrogenation is a combination of the ability of the catalyst for the reverse water-gas shift reaction (CO<sub>2</sub> + H<sub>2</sub> → CO + H<sub>2</sub>O) and the carbon monoxide hydrogenation to hydrocarbons (CO + H<sub>2</sub> → –CH<sub>2</sub>– + H<sub>2</sub>O). For instance, the sample 95Fe-Cu, which displayed a high activity in the CO hydrogenation reaction (Table 2), lacked of any activity in the formation of hydrocarbons from CO<sub>2</sub> hydrogenation because it is not able to convert CO<sub>2</sub> into CO *via* the reverse water-gas shift reaction. On the contrary, K-promoted samples (95Fe-K and 95Fe-CuK) were active in both reverse water-gas shift reaction and CO hydrogenation. Therefore, such catalysts would be particularly suited for the production of hydrocarbons from CO<sub>2</sub>. Fig. 9 shows the hydrocarbons productivity from CO and CO<sub>2</sub> hydrogenation, as well as the approach to the WGS equilibrium for the different catalysts. As it can be seen, the potassium-promoted samples presented the highest activity for WGS reaction and hydrocarbon formation.

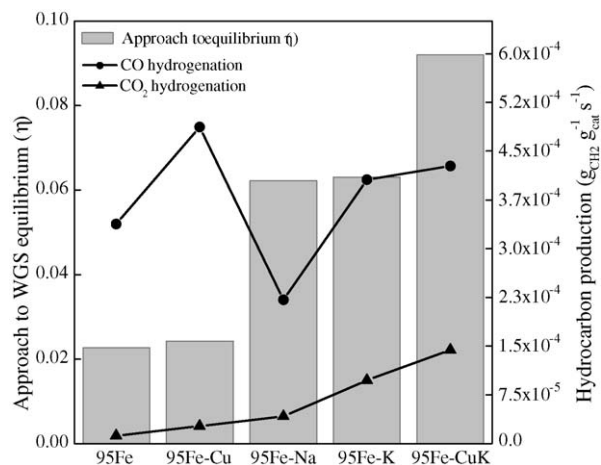


Fig. 9. Approach to the WGS equilibrium and hydrocarbon production in CO and CO<sub>2</sub> hydrogenation at 573 K.

#### 4. Conclusions

The microemulsion technology renders homogeneous Fe–Mn mixed oxides that were afterwards promoted with copper, sodium and potassium. The hydrogenation of carbon monoxide and carbon dioxide was performed with these materials. Although carbon oxides conversion level was similar over iron and iron–manganese catalysts, the latter yielded a higher activity towards hydrocarbons formation. Alkaline promoters (K and Na) increased C<sub>5+</sub> selectivity and CO<sub>2</sub> conversion. Besides, promotion with potassium was more effective, increasing significantly not only hydrocarbon production but also selectivity toward olefins and long-chain products. The addition of copper improved both carbon monoxide and carbon dioxide conversion without modifying products selectivity.

In order to hydrogenate CO<sub>2</sub>, catalysts should be active in both water-gas shift and CO hydrogenation reaction.

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#### References

- [1] M.E. Dry, *Appl. Catal. A Gen.* 138 (1996) 319.
- [2] J.R. Rostrup-Nielsen, *Catal. Rev. Sci. Eng.* 46 (2004) 247.
- [3] D.J. Wilhelm, D.R. Simbeck, A.D. Karp, R.L. Dickenson, *Fuel Process. Technol.* 71 (2001) 139.
- [4] P. Chiesa, S. Consonni, Second Annual Conference on Carbon Sequestration, Washington, 2003.
- [5] T. Riedel, M. Claeys, H. Schulz, G. Schaub, S.S. Nam, K.W. Jun, M.J. Choi, G. Kishan, K.W. Lee, *Appl. Catal. A Gen.* 186 (1999) 201.
- [6] J.F. Lee, W.S. Chern, M.D. Lee, T.Y. Dong, *Can. J. Chem. Eng.* 70 (1992) 511.
- [7] T. Riedel, G. Schaub, K.W. Jun, K.W. Lee, *Ind. Eng. Chem. Res.* 40 (2001) 1355.
- [8] R.A. Fiato, E. Iglesia, G.W. Rice, S.L. Soled, *Adv. Surf. Sci. Catal.* 117 (1998) 339.
- [9] S. Krishnamoorthy, A.W. Li, E. Iglesia, *Catal. Lett.* 80 (2002) 77.
- [10] G.P. van der Laan, A.A.C.M. Beenackers, *Catal. Rev. Sci. Eng.* 41 (1999) 255.
- [11] H. Schulz, *Appl. Catal. A Gen.* 186 (1999) 3.
- [12] D.S. Newsome, *Catal. Rev. Sci. Eng.* 21 (1980) 275.
- [13] Y. Jin, A. Datye, *J. Catal.* 196 (2000) 8.
- [14] L.G. Xu, S.Q. Bao, R.J. O'Brien, A. Raje, B.H. Davis, *Chemtech* 28 (1998) 47.
- [15] J.P. Hindermann, G.J. Hutchings, A. Kiennemann, *Catal. Rev. Sci. Eng.* 35 (1993) 1.
- [16] C.K. Das, N.S. Das, D.P. Choudhury, G. Ravichandran, D.K. Chakrabarty, *Appl. Catal. A Gen.* 11 (1994) 119.
- [17] R. Malessa, M. Baerns, *Ind. Eng. Chem. Res.* 27 (1988) 279.
- [18] G.C. Maiti, R. Malessa, U. Löncher, H. Papp, M. Baerns, *Appl. Catal.* 16 (1985) 215.
- [19] J. Barraault, C. Forquy, V. Perrichon, *Appl. Catal.* 5 (1983) 119.
- [20] W.L. van Dijk, J.W. Niemantsverdriet, A.M. van der Kraan, H.S. van der Baan, *Appl. Catal.* 2 (1982) 273.
- [21] L.Y. Xu, Q.X. Wang, Y.D. Xu, J.S. Huang, *Catal. Lett.* 31 (1995) 253.
- [22] I.R. Leith, M.G. Howden, *Appl. Catal.* 37 (1988) 75.
- [23] G.J. Hutchings, J.C.A. Boeyens, *J. Catal.* 100 (1986) 507.
- [24] Y. Yang, H.W. Xiang, Y.Y. Xu, L. Bai, Y.W. Li, *Appl. Catal. A Gen.* 266 (2004) 181.
- [25] R.B. Anderson, in: Paul H. Emmet (Ed.), *Catalysis*, vol. IV, Academic Press, Orlando, 1956, p. 2.
- [26] Y.Y. Ji, H.W. Xiang, J.L. Yang, Y.Y. Xu, Y.W. Li, B. Zhong, *Appl. Catal. A Gen.* 214 (2001) 77.
- [27] D.C. Silverman, M. Boudart, *J. Catal.* 77 (1982) 208.
- [28] H.W. Pennline, M.F. Zarochak, J.M. Stencil, J.R. Diehl, *Ind. Eng. Chem. Res.* 26 (1987) 595.
- [29] S. Li, A. Li, S. Krishnamoorthy, E. Iglesia, *Catal. Lett.* 77 (2001) 197.
- [30] M.J. Schwuger, K. Stickdorn, *Chem. Rev.* 95 (1995) 849.
- [31] M.A. Lopez-Quintela, *Curr. Opin. Colloid Interface Sci.* 8 (2003) 137.
- [32] S. Eriksson, U. Nylén, S. Rojas, M. Boutonnet, *Appl. Catal. A Gen.* 265 (2004) 207.
- [33] H. Hayashi, L.Z. Chen, T. Tago, M. Kishida, K. Wakabayashi, *Appl. Catal. A Gen.* 231 (2002) 81.
- [34] C. Liu, B. Zou, A.J. Rondinone, Z.J. Zhang, *J. Phys. Chem. B* 104 (2000) 1141.
- [35] Y. Lee, J. Lee, C.J. Bae, J.G. Park, H.J. Noh, J.H. Park, T. Hyeon, *Adv. Funct. Mater.* 15 (2005) 503.
- [36] T. Herranz, S. Rojas, F.J. Perez-Alonso, M. Ojeda, P. Terreros, J.L.G. Fierro, K. Pirota, *Chem. Mater.* 18 (2006) 2364.
- [37] S. Rojas, S. Eriksson, M. Boutonnet, *Catalysis* 17 (2004) 258.
- [38] M.J. Schwuger, K. Stickdorn, R. Schomäcker, *Chem. Rev.* 95 (1995) 849.
- [39] S. Li, W. Ding, G.D. Meizner, E. Iglesia, *J. Phys. Chem. B* 106 (2002) 85.
- [40] R.M. Cornell, U. Schwertmann, *The Iron Oxides: Structure, Properties, Reactions and Uses*, VCH Publishers, New York, 1996.
- [41] N.K. Jaggi, L.H. Schwartz, J.B. Butt, H. Papp, M. Baerns, *Appl. Catal.* 13 (1985) 347.
- [42] G.C. Maiti, R. Malessa, M. Baerns, *Appl. Catal.* 5 (1983) 151.
- [43] S.L. Gonzalez-Cortes, S.M.A. Rodulfo-Baechler, A. Oliveiros, J. Orozco, B. Fontal, A.J. Mora, J. Delgado, *React. Kinet. Catal. Lett.* 75 (2002) 3.
- [44] R.M. Cornell, R. Giovanoli, *Clays Clay Miner.* 35 (1987) 11.
- [45] D.R. Milburn, K.V.R. Chary, B.H. Davis, *Appl. Catal. A Gen.* 144 (1996) 121.
- [46] J. Goldstein, *Scanning Electron Microscopy and X-Ray Microanalysis*, Plenum Press, New York, 1981.
- [47] S.A. Eliason, C.H. Bartholomew, *Appl. Catal. A Gen.* 186 (1999) 229.
- [48] S. Li, G.D. Meizner, E. Iglesia, *J. Phys. Chem. B* 105 (2001) 5743.
- [49] S.T. Oyama, J.C. Schlatter, J.E. Metcalfe III, J.M. Lambert Jr., *Ind. Eng. Chem. Res.* 27 (1988) 1639.
- [50] J. Barraault, C. Renard, *Appl. Catal.* 14 (1985) 133.
- [51] X. Li, B. Zhong, S. Peng, Q. Wang, *Catal. Lett.* 23 (1994) 245.
- [52] A.P. Raje, R.J. O'Brien, B.H. Davis, *J. Catal.* 180 (1998) 36.
- [53] G.J. Hutchings, F.M. Gottschalk, *Appl. Catal.* 51 (1989) 127.
- [54] G.A. Huff, C.N. Sattelfield, *Ind. Eng. Chem. Proc. Des. Dev.* 23 (1984) 696.
- [55] A.R. Suzdorf, N.I. Bakhmutova, A.A. Efremov, L.I. Kuznetsova, B.N. Kuznetsov, *React. Kinet. Catal. Lett.* 21 (1986) 343.